Housing Price Prediction

The house price prediction explains the cost of the building with the features included. They depend on the factors such as the size of the house, the rooms included, nearby area developments. Here we train the dataset using machine learning algorithms and estimate the price according to the data available.

Problem statement:

Buying and selling a house involves the selling price to be correctly estimated. The project determines the house price predicted based on the features.

Project Objectives:

- Prediction of house price
- Data preprocessing to improve data quality
- Analyze the dataset using EDA
- Build a model
- Evaluate the performance
- Deploy the trained model for prediction

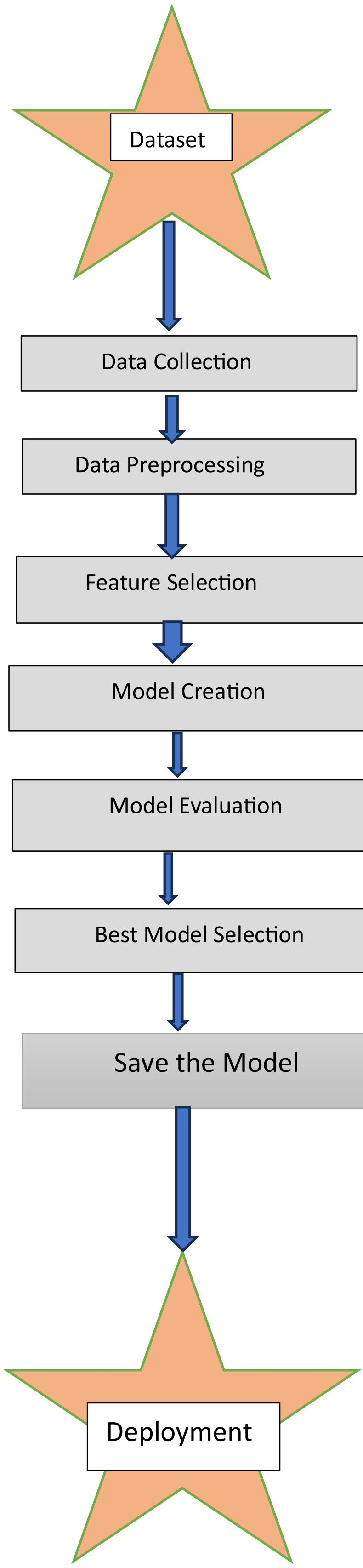
Dataset Description:

The dataset contains information about different houses.

Features:

SNO	Feature	Description
1	Square Feet	Total area of the house
2	Bedrooms	Number of bedrooms
3	Bathrooms	Number of bathrooms
4	Garage Spaces	Number of garages
5	Lot Size	Total land area
6	Nearby School Rating	School rating near the property
7	Crime Rate Index	Crime index of the area
8	Days on Market	Number of days the house was listed
9	Property Age	Age of the property
10	House Price	Target variable

Work flow:


Dataset
Data Collection
Data Preprocessing
Feature Selection
Model Creation
Model Evaluation
Best Model Selection
Save the Model
Deployment

Data Collection:

The housing dataset was loaded into a Pandas DataFrame. The dataset was inspected by checking:

- Number of rows and columns
- Data types
- Column names
- First five records

This helped understand the dataset before preprocessing.

Data Preprocessing

The preprocessing stage improved the quality of the dataset before training.

The following tasks were performed:

- Checked dataset dimensions
- Checked data types
- Identified missing values
- Filled missing numerical values using the median
- Identified duplicate records
- Removed duplicate records
- Detected outliers using boxplots and the IQR method
- Saved the cleaned dataset as **preprocessed_data.csv**

Exploratory Data Analysis (EDA)

EDA was performed to understand the data and identify relationships between variables.

Univariate Analysis

The following analyses were performed:

- Distribution of House Price
- Distribution of Bedrooms

- Distribution of Bathrooms
- Distribution of Square Feet
- Summary statistics
- Boxplots for numerical variables

Purpose

To understand the distribution and characteristics of individual variables.

Bivariate Analysis

Relationships between two variables were analyzed using:

- House Price vs Square Feet
- House Price vs Bedrooms
- House Price vs Bathrooms
- Correlation Heatmap
- Pair Plot

Purpose

To identify the features that have the greatest influence on house price.

Feature Selection

The correlation matrix was used to identify important numerical features.

Highly correlated variables with house price were selected as input features for the Linear Regression model.

Model Creation

The cleaned dataset was divided into training and testing datasets using the train-test split method.

A Linear Regression model was trained using the training dataset.

Model Evaluation

The model was evaluated using the following metrics:

- Mean Absolute Error (MAE)

- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

These metrics measure the prediction accuracy of the model.

Model Saving

The trained model was saved.

The saved model can be loaded later without retraining.

Deployment

The saved model was deployed using Python.

Users enter house details such as:

- Square Feet
- Bedrooms
- Bathrooms
- Garage Spaces
- School Rating
- Crime Rate

The application predicts the estimated house price using the trained Linear Regression model.

Technologies Used

Technology	**Purpose**
Python	Programming language
Pandas	Data manipulation
NumPy	Numerical computation
Matplotlib	Data visualization

Technology	Purpose
Seaborn	Statistical visualization
Scikit-learn	Machine learning
Pickle dump	Save and load model
Jupyter Notebook	Development environment

Conclusion

This project successfully predicts house prices using a Linear Regression model. Data preprocessing and exploratory data analysis improved data quality and provided insights into the factors affecting house prices. The trained model can estimate house prices based on user inputs and demonstrates a complete machine learning workflow from data collection to deployment.